

* Inner or Nested loops :-

- the process of defining one loop inside of another loop is called Inner or Nested loop.
- the execution process of inner loops is that "for every value of outer loop, Inner loop repeated for finite number of times until Inner loop condition becomes 'false'"
- In python programming, we can use Inner loops in 4 approaches. They are:

approach ① :- for loop in for loop

```
for varname1 in iterable-object1 : # outloop
    for varname2 in iterable-object2 : # innerloop
        _ _ _ _ _
    else:
        _ _ _ _ _
else:
    _ _ _ _ _
```

approach ② while loop in while loop

```
while (Test Cond1) : # outloop
    while (Test Cond 2) : # innerloop
        _ _ _ _ _
    else:
        _ _ _ _ _
else:
    _ _ _ _ _
```

approaches ③ :- for loop in while loop

```
while (Test Cond1) : # outloop
    for varname2 in iterable-object2 : # innerloop
        _ _ _ _ _
    else:
        _ _ _ _ _
else:
    _ _ _ _ _
```

approaches ④ :- while loop in for loop

```
for varname in iterable-object1 : # outloop
    while (Test Cond) : # outloop
        while (Test Cond 2) : # inner loop
            _ _ _ _ _
        else:
            _ _ _ _ _
    else:
        _ _ _ _ _
```